

Jing-Zhang Smart Vein · Boundless Green

百年京张 AI 创新带城市设计方案 (A3 Design Booklet)

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Centennial Jing-Zhang AI Innovation Belt Urban Design: Smart Axis & Unbounded Green

(详见 proposal.md 正文)

Design Basis and Source List

This formal proposal takes the *Call for Prequalification for the International Urban Design Solicitation of the Centennial Jing-Zhang AI Innovation Belt* published by the Haidian Branch of the Beijing Municipal Commission of Planning and Natural Resources as its primary basis, with the provisional boundaries, key areas, enums, ranges and source registry maintained in `brief/site-package/` as its machine-readable basis. Before generation, the AI agent fully read `design\_brief.json`, `allowed\_design\_space.json`, `sources.json`, `enums/`, `ranges/`, `schemas/`, `data/source\_registry.json` and `data/processed/agent\_fact\_pack.md`, and built task, scope, source-use and gap checklists. The announcement requires urban-design depth at the regulatory detailed planning level and at the comprehensive implementation planning level, so the narrative is tightly coupled with GeoJSON layers, indicator t

Three-Level Scope Framework

The proposal follows the three tiers defined by the announcement: 1. **Regional study scope (43.6 km²)**: the AI industry ecosystem, strategic positioning, innovation chain and future urban form of the southern Haidian innovation corridor, bounded by the North 5th Ring Road to the north, the Jingzang Expressway to the east, Xizhimen Outer Street to the south, and Wanquanhe Road to the west. 2. **Overall design scope (11.4 km²)**: the 1-2 km urban belt around the Jing-Zhang Railway Heritage Park, forming the overall renewal framework, industrial spatial layout, blue-green slow-traffic network, transport/municipal support and urban-character control. 3. **Key area scope (368.4 ha)**: three core districts designed in detail from north to south — Zhongzhiyuan AI Self-Innovation Acceleration Area (192.1 ha), Beijing AI Origin Community (104.3 ha), and Dazhongsi AI Industry Cluster (72.0 ha).

Coordinated Research Area: Industry and Future City Research

Global Benchmarks & Regional Synergy Six global AI ecosystem benchmarks: 1. **San Francisco Mission Bay**: university-industry-VC clustering, applied to the Origin Community. 2. **London Tech City**: high-density renewal and creative-industry integration, empowering Dazhongsi. 3. **Toronto MaRS Discovery District**: innovation-hub shared research resources, guiding Zhongzhiyuan. 4. **New York Cornell Tech**: open campus with corporate test sandboxes, strengthening industry-academia links. 5. **Boston Kendall Square**: high walkability and dense innovation networks, optimizing the blue-green slow system. 6. **Singapore Punggol Digital District**: system-level connectivity, distributed sensing, micro-grids and smart environmental governance. **Three-areas-two-wings synergy and Beijing-Tianjin-Hebei linkage**: beyond the three cores, the proposal builds a "two-wings" system: westward, t

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

The overall design scope (11.4 km²) reaches regulatory detailed planning urban-design depth, satisfying [depth:land_use_layout] and [depth:development_intensity_controls]. Four functional zones are planned in `geometry/land\_use.geojson`: - **LU-001: AI R&D; innovation and full-stack self-innovation land (311.9 ha)**: model R&D;, algorithm research and software-hardware synergy, located in the northern Zhongzhiyuan area. - **LU-002: Jing-Zhang unbounded green park and open space (270.8 ha)**: a north-south green belt and waterfront park system formed by the linear spine park, pocket parks and riparian green space, reaching a 31.1% green ratio. - **LU-003: AI industry services and commercial headquarters land (309.4 ha)**: headquarters, an international demo hub and fintech services in the southern Dazhongsi area. - **LU-004: AI origin international talent community and quality living land (

Detailed Design of Key Areas

The key areas cover 368.4 ha in three districts, responding to [depth:three_key_area_detailed_design]: ### 1. Zhongzhiyuan AI Acceleration Area (192.1 ha) - **Positioning**: garden-type full-stack self-innovation district. - **Spatial moves**: a low-carbon innovation waterfront along the Qinghe River; a national AI model test ground and software-hardware compatibility validation lab. - **AI scenarios**: concept-planning "02 Safety-Governance Sandbox" and "06 Qinghe Low-Carbon Innovation Waterfront" with low-carbon compute and standards workshops. - **References**: [data:geometry/key_areas.geojson#PROV-KEY-001], [depth:three_key_area_detailed_design]. ### 2. Beijing AI Origin Community (104.3 ha) - **Positioning**: campus-adjacent commercialization and youth-talent community. - **Spatial moves**: stitching gaps at Qinghua East Road West and Wudaokou between campuses and parks; converting

AI Innovation Ecosystem, Personas, and AI+ Scenarios

The proposal defines 6 typical personas and deploys 12 spatial scenario cards plus 3 AI pilgrimage landmarks, per [source:AGENT-TASKBOOK]: ### 6 Personas and need responses 1. **Open-source developers**: deeply dependent on communities, contribution display and nighttime exchange; 24h open-source launch hall and co-working spaces in the Origin Community. 2. **Startups**: low-cost offices, compute subsidies and test access; shared edge-compute stations and model red-team sandboxes in Zhongzhiyuan. 3. **Headquarters visitors**: international display, business meetings and efficient transit; international demo lounge and TOD links at Dazhongsi. 4. **Residents**: park leisure, community services and low-disruption renewal; slow green loop and embedded smart community service points along the heritage park. 5. **University faculty and students**: campus-adjacent commercialization and cross-ca

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Fine-grained Retain-Renovate-Demolish Principles Based on Existing Park Given that Jing-Zhang Railway Heritage Park Phase I has been completed and opened, and Phase II is underway ([source:OFFICIAL-ANNOUNCEMENT], the announcement references "existing implemented areas of the park"), this proposal strictly adheres to a fact-based approach: it does not redraw or alter the completed park landscaping, but treats it as a static "base layer" to be fully preserved, with AI scenarios and smart facilities overlaid on top. All spatial actions are precisely focused on "stitching slow-mobility breakpoints on both sides of the park" (e.g., Qinghua East Road West Entrance, Wudaokou intersections), "implanting edge-computing supply nodes," and "activating existing factory buildings along the line as near-campus transformation spaces," achieving a seamless weave of historical-cultural, urban-life, a

Transport, Rail, Municipal Infrastructure, and Public Services

Mobility and municipal planning follow [depth:traffic_rail_slow_parking] and [depth:municipal_new_infrastructure]: ### 1. Transit & TOD - **Station stitching**: improve integration at Wudaokou, Qinghua East Road West and Dazhongsi stations; explore a 15-minute rail-walk circle. - **Elevated and underground links**: explore three-dimensional links at Dazhongsi and Wudaokou to ease vehicle-pedestrian conflicts (pending municipal engineering feasibility). ### 2. Slow traffic & green mobility - **Continuous slow greenway**: a continuous cycling and walking trunk along the Heritage Park crossing major intersections (alignment and length pending professional verification). - **Autonomous micro-circulation**: concept-planning autonomous shuttle loops connecting stations, parks and communities. ### 3. Three test-scenario tiers and admission matrix - **Distributed edge-compute network**: discreet

Blue-Green Network, Public Space, and Urban Character

The blue-green plan follows [depth:blue_green_public_space], using the Heritage Park as the north-south ecological spine with the Qinghe and Xiaoyuehe rivers, forming a "one axis, two rivers, multiple corridors, a hundred parks" pattern. - **Green ratio & public space**: 3.55 km² of park green space, a 31.1% green ratio ([metric:green_ratio]), and a 25.3% public open-space ratio ([metric:public_space_ratio]). - **Urban character**: three material languages — centennial Jing-Zhang industrial red brick, Zhongguancun tech gray aluminum, and future AI clear glass — shaping interfaces that balance heritage and futurity; continuous park greenways connect 12 communities and universities. ### Qinghe Low-Carbon Waterfront Multi-Species Eco-Sensing and Resilience System Concept proposal to explore a multi-species eco-sensing and resilience system at the Qinghe low-carbon waterfront (for profession

Renewal Projects, Implementation Policy, and Phasing

Following [depth:renewal_project_list] and [depth:phasing_implementation], the proposal sets a "visible in three years, exemplary in five, benchmark in ten" phasing plan, with responsible-department suggestions and risk-control measures for major works: ### Long-term operation & governance roadmap - **Privacy & safety sandbox**: data minimization, "usable but invisible" processing and periodic destruction for all deployments; human-in-the-loop posts and digital-exclusion compensation services. - **Component library & cultural wayfinding**: unified "Jing-Zhang Smart Vein" visual identity (Logo), heritage-based honor display walls, bilingual barrier-free physical guidance. ### Data gaps, privacy protection & degradation mechanisms - **Known gaps**: official precise redline and key-area polygons are not yet published; all area indicators, spatial anchors, FAR and height controls are scenari

Cultural Narrative: Jing-Zhang Heritage, Zhongguancun Culture & AI New Culture

The cultural narrative is the systematic expression of the "Centennial Jing-Zhang" theme. The proposal organizes Jing-Zhang railway heritage, Zhongguancun innovation spirit and AI new culture into a "past-present-future" three-layer narrative mapped to space (agent.5 of [source:AGENT-TASKBOOK]). ### 1. Jing-Zhang railway heritage resource system Three tiers by protection level and spatial character: - **Heritage-grade nodes**: protected sites such as Qinghuayuan Station Heritage Site serve as narrative origins and spatial anchors, limited to protective display and low-impact adaptation (conceptual; subject to heritage approval). - **Site-grade linear heritage**: the Jing-Zhang Railway Heritage Park and relics form the north-south narrative spine for commemoration, education and slow mobility. - **Memory-grade elements**: station buildings, crossings, bridges and industrial structures alo

AI Scenario Admission, Open Operation & Community Governance

1. Industrial test-scenario admission matrix - **Unmanned delivery test section**: L4+ fleets; closed-road low-peak night data required; exit after two at-fault incidents. - **Open-source algorithm evaluation sandbox**: university teams and filed enterprises; physically isolated offline compute; mandatory code-security and bias review. - **Embodied-intelligence inspection park**: security and environmental robots; whitelist management during non-crowded hours. ### 2. Developer community governance & annual events An "Open-Source Governance Committee", an honor display component library, and a Code Wall in the Origin Community launch hall. An event system of "annual developer conference, quarterly algorithm challenges, monthly geek salons" with an enterprise landing fast track forms the "talent aggregation - idea validation - investment incubation" loop. ### 2.1 City-as-Repo Open Sour

Metrics, Area Recalculation, and Compliance Matrix

All known indicators exactly match GeoJSON features: - **Site area [metric:site_area_sqm]**: about 11.41 km² (11.4 km²) - **Key areas [metric:key_area_count]**: 3 districts, 3,684,000 m² (368.4 ha) - **Building footprint [metric:building_footprint_area_sqm]**: about 1.80 km² - **Green ratio [metric:green_ratio]**: 31.1% (about 3.55 km² / 11.41 km²) - **Public-space ratio [metric:public_space_ratio]**: 25.3% (about 2.89 km² / 11.41 km²) - **FAR [metric:floor_area_ratio]**: to be calculated after official regulatory controls are released - **Compliance response**: fully responds to announcement tasks 1.3, 1.4, 1.5 and agent tasks agent.1 - agent.6.

Risk, Copyright, and Compliance

1. **Provisional boundary declaration**: official polygon is not yet provided; all boundaries and area indicators built on `provisional_boundaries.geojson` are provisional conceptual delineations (`official_boundary=false`) for concept generation and display only, and are not legal redlines or approval conclusions; full recalculation is required once official data is supplied. Qualification, scoring, acceptance, publication, and merging are to be determined by maintainers following trusted validation. 2. **AI generation & compliance review**: generated by an AI agent from public sources and regulations under data-minimization and privacy principles; no non-public confidential data. 3. **Copyright & legal disclaimer**: COMMUNITY-DISPLAY-ONLY license; all images, drawings and code assets are cleared under the open-call rules. All spatial moves and compute deployments are conceptual design

Visual Index

Index of all visual assets, drawings and digital exhibits:

References

- brief/public-brief.md - brief/site-package/design_brief.json - brief/site-package/allowed_design_space.json - brief/site-package/enums/ - brief/site-package/ranges/planning_limits.json - data/processed/agent_fact_pack.md - Machine-readable reference index: [source:OFFICIAL-ANNOUNCEMENT], [source:AGENT-TASKBOOK], [source:SITE-PACKAGE], [source:SOURCE-REGISTRY], [source:PROCESSED-FACT-PACK], [standard:PROJECT-OFFICIAL-ANNOUNCEMENT], [standard:PROJECT-AGENT-OPEN-CALL-TASKBOOK], [depth:metrics_recalculation], [data:geometry/site_boundary.geojson#SITE-001], [metric:site_area_sqm] ### Civic Value Protocol (Civic Compute Revenue Reinvestment) Conceptual proposal to establish the Civic Value Protocol: - **Public Reinvestment Mechanism**: Conceptually suggests reinvesting 15% of operational revenues from commercial AI compute nodes into the Jing-Zhang Civic Value Fund. - **Targeted Community Su